CSE 460
VLSI DESIGN SECTION-10

ASSIGNMENT-2

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1) Derive the Moore type state diagram, state table, state assigned table and the final circuit for a modulo-6 up counter (i.e. a counter that counts 0,1,2,3,4,5,0,1,2,...).

- It should have a 1-bit input w, a clock signal and a multi-bit output z.
- The machine should increase its count whenever w = 1, and hold its previous count otherwise.
- At each counting step the machine should output the value of the count in binary.

2) Design both a moore and mealy machine that detects the following pattern in its input (w) : 1, 0, 1. Draw the state diagram, state table, state assigned table and the final circuit.

- Repeating and overlapping sequences should be detected.
- The machine should generate the output z = 1 if in the past 3 clock cycles 101 pattern was detected, and z = 0 otherwise.

3) An FSM that has an input w and an output z. The machine has to generate z = 1 when the previous four values of w were 1001 or 1111; otherwise, z = 0. Overlapping input patterns are allowed. An example of the desired behavior is:

w : 0101111001100111 11

z : 0000001001000100 11

4)

Consider a sequential circuit that detects if two consecutive bits are the same or not. The circuit generates an output z=1 if two consecutive bits are opposite and z=0 if the bits are the same. The output changes in the same clock cycle where the input is received.

w	0	**0**	**1**	**0**	**0**	**1**	**1**	**0**
z	0	0	**1**	**1**	0	**1**	0	**1**

Consider overlapping bit sequence

(a) **Determine** the type of FSM required for solving this problem. **1**

(b) **Draw** the state diagram of the circuit. **3**

(c) **Derive** a state-assigned table from (b) using the gray-encoding technique. **3**

(d) **Find** the next-state and output logic expressions using k-map. **3**

5)

Consider a sequential circuit with two inputs (p1 & p2) and a single output, z. The decisions regarding the status of various product orders are handled by two departments: the Sales Team and the Warehouse. Both departments independently approve (denoted by 1) or reject (denoted by 0) an order. The decisions of the Sales Team are given by a bit stream, p1, while the decisions of the Warehouse are represented by p2. If, for two consecutive orders, both departments approve (**i.e., p1=p2=1 for two consecutive clock cycles**), the sequential circuit provides **z=1 at the next order**. The decisions of the Sales Team & the Warehouse for several orders and their corresponding synchronization instances are given below:

p1	0	**1**	**1**	**1**	**1**	1	1	0	0	0	1
p2	1	**1**	**1**	**1**	**1**	0	1	0	0	1	1
z	0	0	0	**1**	0	**1**	0	0	0	0	0

Note:

- **You must use binary encoding to assign states.**
- **The overlapping of the input bits (p1 & p2) is not allowed.**

(a) **Determine** the type of FSM that would be needed for the given problem? **1**

(b) **Design** the state diagram for the corresponding FSM **2**

(c) **Derive** the state table and state-assigned table from your state diagram in (b). **3**

(d) **Evaluate** the next state & output logic expressions to implement the FSM & calculate the number of flip-flops required for the circuit implementation. (You don't need to draw the circuit implementation). **4**